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META-ANALYSIS

The role of antibiotic exposure as risk factor for IBD epidemic: an updated meta-analysis

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Key words
antibiotic exposure, Crohn's disease, inflammatory bowel disease, meta-analysis, ulcerative colitis.

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Abstract

Background and Aim: The rising prevalence of IBD globally has raised concerns about antibiotic exposure. This study's meta-analysis examines antibiotic exposure, frequency, year before diagnosis, regional differences, and IBD incidence.
Methods: The literature review used PubMed, Web of Science, Elsevier, ScienceDirect, and Cochrane CENTRAL databases up to June 2024 to explore the link between antibiotic exposure and IBD risk. Stratified analysis was conducted by years of antibiotic exposure before IBD diagnosis, frequency, and region. Adjusted odds ratios (ORs) and 95% confidence intervals (CIs) were combined using a random effects model.
Results: Eighteen case–control studies and five cohort studies were included (n = 99, 104 IBD patients and n = 2 273 336 controls). The findings indicate that antibiotic exposure significantly has a positive association with the risk of developing IBD (OR, 1.66; 95% CI, 1.28–2.16). Antibiotic exposure of ≥3 years (OR, 1.49; 95% CI, 1.12–1.98), 2 years (OR, 1.46; 95% CI, 1.37–1.55), and ≤1 year (OR, 1.55; 95% CI, 1.17–2.04) prior to the diagnosis of IBD is associated with a higher risk of developing IBD. Cumulative exposure of ≥3 dispensations (OR, 2.02; 95% CI, 1.49–2.74) and two dispensations (OR, 1.36; 95% CI, 1.03–1.78) also had a positive association with IBD risk, while one dispensation did not (OR, 0.96; 95% CI, 0.72–1.26). No significant association was found in developing countries (OR, 1.92; 95% CI, 0.71–5.19), but developed countries showed a significant positive association with the risk (OR, 1.58; 95% CI, 1.27–1.96).
Conclusion: The meta-analysis suggests that antibiotic use has a positive association with the risk of IBD, and limiting unnecessary antibiotic use may be one way to reduce the risk of developing IBD.

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Introduction

Inflammatory bowel disease (IBD) is a chronic inflammatory condition of the gastrointestinal tract, encompassing Crohn's disease (CD) and ulcerative colitis (UC). It primarily affects the rectum and colon. Clinical manifestations typically include bloody diarrhea, abdominal pain, fever, and anemia. As of 2023, the global incidence of UC alone was approximately 5 million cases, with the worldwide incidence on the rise.[1]

Recent studies indicate that antibiotic usage can significantly disrupt gut microbiota, leading to microbial imbalance, or dysbiosis. This disruption provokes immune responses, culminating in gastrointestinal irritation that may elevate the risk of IBD and accelerate its progression. Evidence suggests that excessive antibiotic use can cause lasting damage to gut microbiota.[2–4] Consequently, there is increasing public concern about the potential link between antibiotic exposure and heightened IBD risk.

The initial meta-analysis investigating the correlation between antibiotic exposure and IBD was carried out by Ungaro *et al*. in 2014.[5] Subsequently, a substantial body of literature has continued to emerge on this topic, including several large-scale national case–control and cohort studies. Furthermore, previous meta-analyses did not consider the relationship between cumulative antibiotic exposure frequency, years of antibiotic exposure before IBD diagnosis, and regional differences on the risk of developing IBD. Therefore, we conducted an updated systematic review and meta-analysis to reevaluate the association between

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antibiotic exposure and IBD, taking into account specific antibiotic usage frequencies and regional differences.

Methods

According to Hsu *et al*. and Hwang and Tsai,[6,7] it is crucial to conduct a review based on high-quality publications. This meta-analysis was conducted following the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)[8] guidelines. All steps of the literature search, study selection, quality assessment, and data extraction were performed independently by three authors (Rui pan, Yang He, and Shujie Zhao) using a standardized protocol. All the statistical analyses and graphics were performed with the R 4.4.0 statistical package (R version 4.4.0 Puppy Cup). Any discrepancies were resolved through discussion, leading to consensus.

Literature search. We conducted an electronic literature search using PubMed, Web of Science, Elsevier, ScienceDirect, and Cochrane CENTRAL databases, covering publications up to June 2024. The search terms used in PubMed, Web of Science, and Elsevier, ScienceDirect databases included ("antimicrobial" OR "antibiotic" OR "bactericidal" OR "anti-bacterial" OR "anti-bacterials" OR "anti-bacterial agent" OR "anti-bacterial agents" OR "bactericidal compound" OR "bactericidal compounds" OR "bactericidal agent" OR "penicillin" OR "cephalosporin" OR "tetracycline" OR "doxycycline" OR "minocycline" OR "fluoroquinolone" OR "macrolide" OR "sulfonamide" OR "metronidazole" OR "aminoglycosides" OR "vancomycin") AND ("inflammatory bowel disease" OR "Crohn" OR "ulcerative colitis" OR "UC" OR "CD" OR "inflammatory bowel diseases") AND ("risk" OR "Association" OR "relationship" OR "danger"). Cochrane CENTRAL databases were searched using the following search terms ("inflammatory bowel disease" OR "Crohn" OR "ulcerative colitis" OR "IBD") AND ("antimicrobial" OR "antibiotic" OR "Anti-Bacterial Agents").

Inclusion and exclusion criteria. To ensure the quality of the literature utilized in this study, specific inclusion and exclusion criteria were established (Table 1). Each paper was carefully screened to determine its suitability for analysis. Only studies discussing the relationship between antibiotic exposure and IBD, containing relative risk (RR) or odds ratio (OR) with 95% confidence intervals (95% CI) in their results, and providing key information such as the exact number of individuals exposed to antibiotics and the number of IBD cases were considered.

Quality assessment. The studies included in the analysis consisted of cohort and case–control studies. The quality of these studies was evaluated using the Newcastle-Ottawa Scale (NOS)[9] on a scale from 0 to 9. The quality standards are defined as high quality (≥7), medium quality (4-6), and low quality (≤3).

Result

Literature search results. According to the PRISMA review process, a total of 3476 articles were retrieved from the

Table 1 Inclusion and exclusion criteria

Inclusion criteria		Exclusion criteria	
1	Original studies that explored the association between antibiotic exposure and the risk of new-onset IBD	1	Literature reviews, commentaries, or meta-analysis
2	The paper includes specific numbers of antibiotic exposures and incidence rates	2	The paper does not contain specific data
3	Observational research based on people	3	Animal and in vitro experiments
4	Written in English	4	Written in other languages

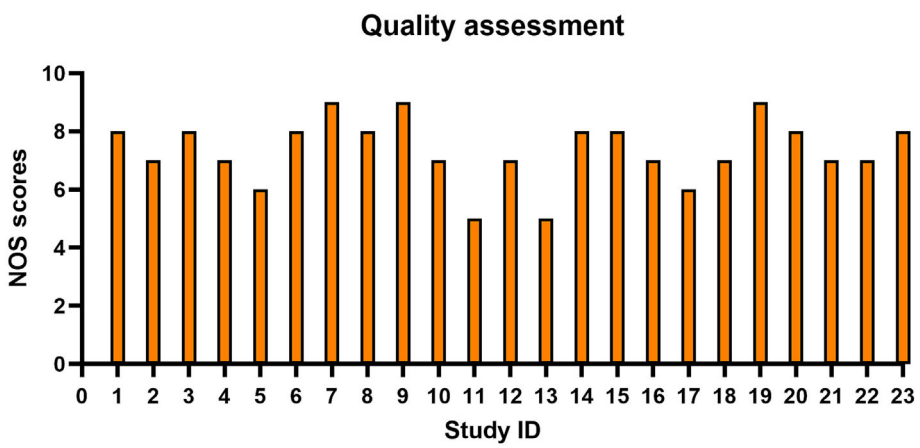


Figure 1 Quality assessment.

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Table 2 Characteristics of included case–control studies

Study ID	Author	Year	Country	Cases	Controls	Age of population (mean, SD)	Excluded time of exposure before IBD diagnosis	Outcomes were adjusted for the following variables
1	Oh *et al.*[10]	2023	Korea	60 764 IBD 49 384 UC 10 854 CD	303 820	45.2 (16.8)	2–5 years	Age, sex, residence, body mass index, smoking status
2	Shimodaira *et al.*[11]	2022	Japan	2791 IBD 371 CD 2420 UC	13 955	39.4 (13.5)	3–12 months	Time, age, sex, presence of enterocolitis
3	Mak *et al.*[12]	2022	Australia, China	254 CD	78	38.57 (14.08)	Before 18 years	NR
4	Lee *et al.*[13]	2022	Malaysia	70 IBD 38 CD 32 UC	140	7.0	NR	Gender, ethnicity, age
5	Nguyen *et al.*[14]	2020	Sweden	23 982 IBD 7898 CD 15 951 UC	117 827	35	1 years	Age, sex, calendar year, county of residence, number of inpatient and outpatient encounters, education level
6	Canova *et al.*[15]	2020	Italy	70 IBD 38 CD 27 UC	700	8.8 (4.7)	In the first year of life	Sex, year of birth, season of birth, having older siblings, number of births, birth weight, gestational age, Apgar scores at 1 min, maternal age, mother's for maleducation at the moment of birth, gastrointestinal infections
7	Troelsen *et al.*[16]	2020	UK	683 CD 461 UC	4576	NR	1–9 years	Age, sex, general practice, registration year in the CPRD GOLD
8	Aniwan *et al.*[17]	2018	USA	736 IBD 339 CD	1472	34	3 months to 5 years	Age, sex, smoking status, family history of IBD, obesity, statin use
9	Strisciuglio *et al.*[18]	2017	Italy	102 CD	203	11	3–12 months	NR
10	Niu *et al.*[19]	2016	China	102 CD 678 UC	3122	NR	1–2 times/years or ≥3 times/years during childhood	Age, sex
11	Ng *et al.*[20]	2015	Asia-Pacific region	442 IBD 186 CD 256 UC	940	38	>4 times per years in childhood	Age, gender, ethnicity, geographic location
12	Virta *et al.*[21]	2012	Finland	595 IBD 233 CD 362 UC	2380	NR	2 years	Age, sex, additional chronic diseases
13	Castiglione *et al.*[22]	2012	Italy	468 CD 527 UC	562	CD 36 UC 37	During childhood	NR
14	Shaw *et al.*[23]	2011	Canada	2234 IBD 1025 CD 1218 UC	22,346	43.4	2–5 years	Sex, region of residence, age, non-antibiotic prescription patterns, gastrointestinal-related visits
15	Shaw *et al.*[24]	2010	Canada	36 IBD 9 UC 27 CD	360	8.2 (2.7)	1 year	Age, sex, residence area
16	Han *et al.*[25]	2010	New Zealand	315 CD	536	45.6 (0.9)	Developing IBD, in <6 years, 6–11 years, and 12–16 years of age	NR
17	Hildebrand *et al.*[26]	2008	Sweden	590 CD	6550	13.6	NR	Maternal age
18	Card *et al.*[27]	2004	UK	587 CD	1460	43.76	2–5 years	Age, sex, smoking, prescription of other drugs

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databases (PubMed = 764; Elsevier, ScienceDirect = 302; Web of Science = 1993; Cochrane Library = 417). Duplicate articles were removed, and then selected 53 articles were selected based on reading the titles and abstracts. After carefully reviewing the full text and applying the inclusion and exclusion criteria, we included 23 articles,[10–32] as shown in Table S1. The detailed PRISMA review process is depicted in Figure S1.

Quality assessment. The NOS scores of the 23 included studies ranged from 5 to 9, indicating a medium to high quality. This process was jointly completed by three researchers, and the results of the quality assessment are shown in Figure 1.

Study characteristics. This study involved a total of 99 104 IBD patients and 2273 336 controls. The general characteristics of the case–control studies included in this research are shown in Table 2, while the general characteristics of the cohort studies are shown in Table 3. Out of the 23 studies included, 18 were case–control studies, and five were cohort studies. All included studies were published between 2004 and 2023. Notably, there has been a significant increase in publications in the last 5 years, with 10 articles being published during this recent period.

The included studies were mainly from the United Kingdom, the United States, Korea, Canada, Finland, New Zealand, China, Sweden, Italy, and Japan. According to the Human Development Report (UNDP) and World Bank Country and Lending Groups, we classified China, Malaysia, and Indonesia as developing countries. Although some regions in China are economically developed, the studies included in the literature mostly originated from economically less developed areas such as Yunnan. Among the studies, 19 articles investigated the risk of IBD associated with antibiotic exposure in developed countries, and only one research[12] involved both developed and developing countries. Eight studies investigated the impact of exposure to specific antibiotics on the incidence of IBD. According to the general characteristics of these populations, we conducted corresponding meta-analysis and subgroup analysis.

Outcomes

IBD overall. First, the data underwent a pooled analysis. The results revealed an I^2 value greater than 50%, so the random effects model was selected for analysis. The findings showed a significant association between antibiotic exposure and the risk of developing IBD in the population (OR, 1.66; 95% CI, 1.28–2.16) (Fig. 2). In addition, according to the funnel plot results, there was no significant publication bias among the literature included in this study ($P > 0.05$, $SE = 239.6$) (Fig. 3). Subsequently, the sensitivity analysis of the studies included in this meta-analysis, showed that no single study had a significant impact on the results of this meta-analysis (Fig. S2). Next, subgroup analysis of the data was performed (Table 4).

UC. Subgroup analysis of UC patients stratified by age of onset showed no significant association between antibiotic exposure and UC risk in adults (OR, 1.36; 95% CI, 0.95–1.95) (Fig. S3).

Table 3 Characteristics of included cohort studies

Study ID	Author	Year	Country	Cases	Cohort size	Age of population (mean, SD)	Excluded time of exposure before IBD diagnosis	Outcomes were adjusted for the following variables
19	Taylor *et al.*[28]	2023	USA	382	289 697	24.4 (9.9)	1 years	Age, sex, race, ethnicity, combined oral contraceptive
20	Narula *et al.*[29]	2023	24 countries	571 IBD 143 CD 428 UC	133 137	50.3 (9.7)	1 months	Sex, center, smoking, BMI, education, location, Alternative Healthy Eating Index score, diabetes, hypertension, stroke, angina/heart attack/coronary artery disease, heart failure, other heart disease, hepatitis/jaundice, cancer, chronic obstructive pulmonary disease, asthma, tuberculosis
21	Mark-Christensen *et al.*[30]	2022	Denmark	208 CD	979 039	NR	In the first year of life	Route of delivery, maternal age, gestational age, maternal smoking during pregnancy, sex of the infant, degree of urbanization, birth order, NSAID use
22	Lee *et al.*[31]	2013	Australia	18	1167	NR	NR	NR
23	Margolis *et al.*[32]	2010	UK	207 IBD 71 CD 99 UC	94 487	22.1 (5.68)	At least 1 month	Gender, age, practice, use of oral Contraceptive medication (for females), Cigarette use, use of topical medication to treat acne, history of diabetes, history of hypertension

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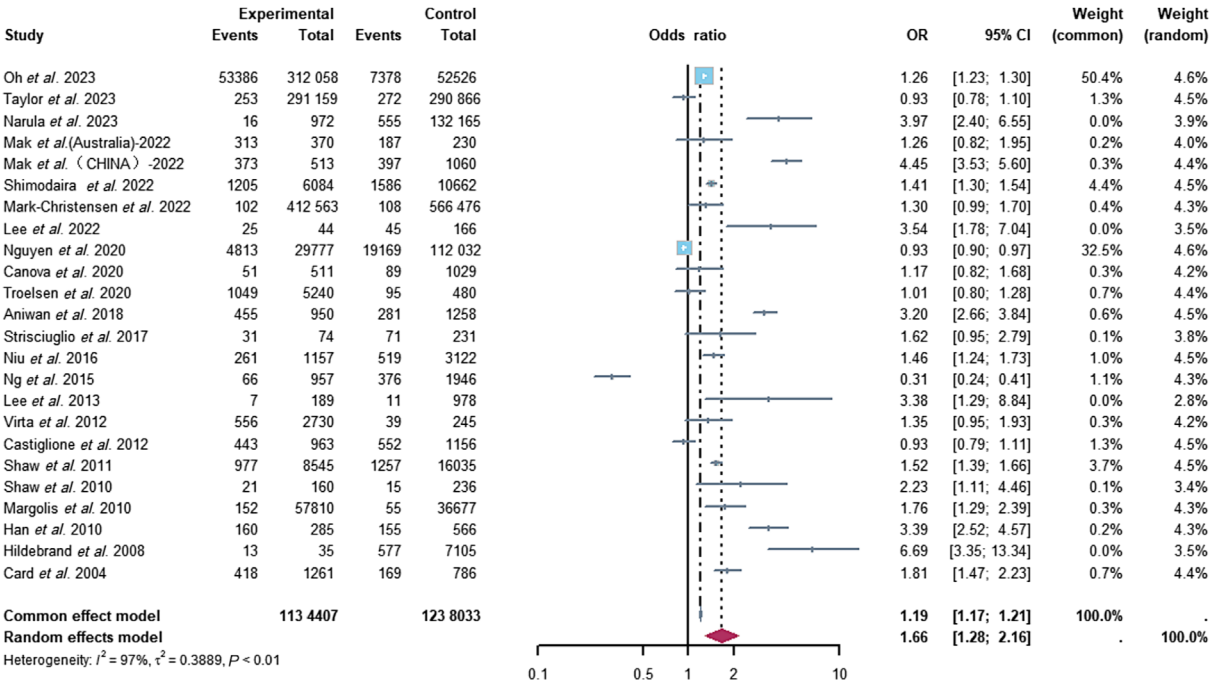


Figure 2 Forest plot summarizing the results of the meta-analysis.

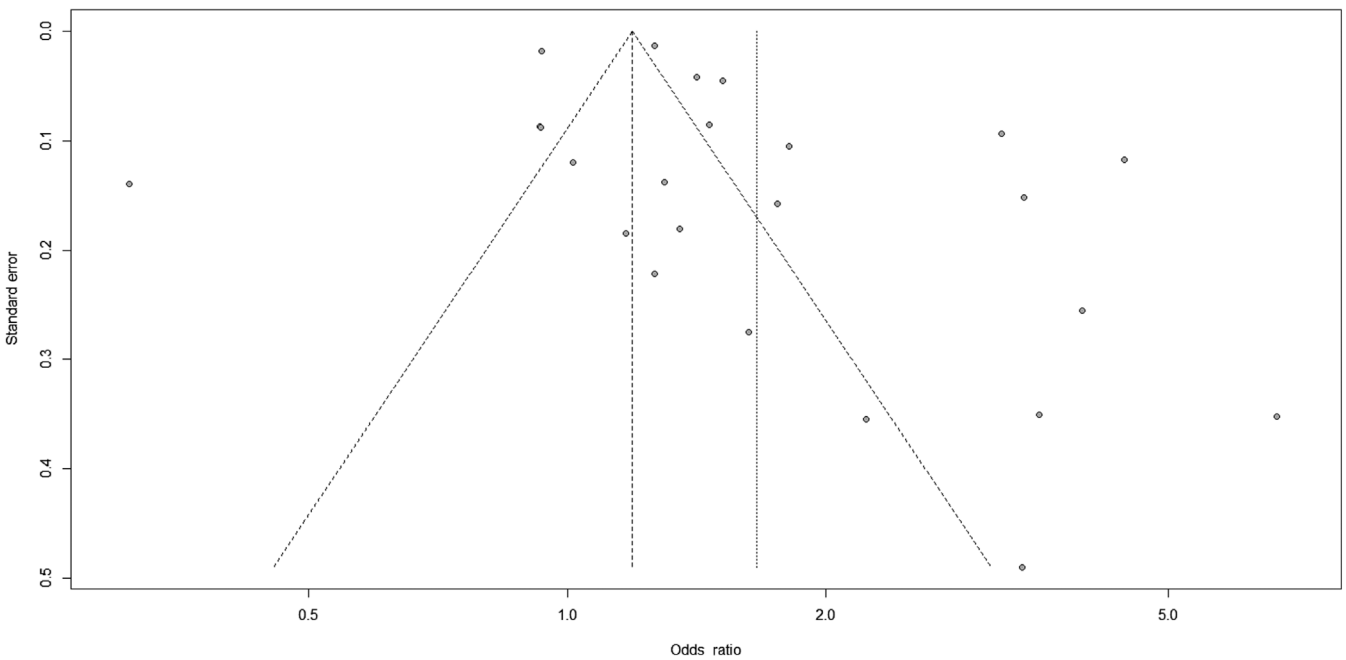


Figure 3 Funnel plot of publication bias for association between antibiotic exposure and risk of new-onset IBD across all studies.

Similarly, there was no significant association between antibiotic exposure and the rick of UC incidence in children (OR, 1.23; 95% CI, 0.94–1.61) (Fig. S4).

CD. For CD patients, subgroup analysis stratified by age at onset revealed a significant association between antibiotic exposure and the incidence risk of adult CD (OR, 1.57; 95% CI, 1.28–1.95)

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Table 4 Subgroup analyses of the association between antibiotic exposure and the risk of new-onset IBD

Subgroup analyses	*n*	Pooled OR (95% CI)	I^2 (%)	*P* value for publication bias
IBD subtype				
Children UC	7	1.23(0.94–1.61)	52%	>0.05
Children CD	11	1.26(0.57–2.76)	96%	>0.05
Adults UC	7	1.36(0.95–1.95)	86%	>0.05
Adults CD	8	1.57(1.26–1.94)	74%	>0.05
Exposure period				
Children	14	1.50(1.03–2.18)	93%	>0.05
Adults	12	1.61(1.22–2.11)	97%	>0.05
Year before diagnosis				
≤1 year	6	1.55(1.17–2.04)	84%	>0.05
2 years	5	1.46(1.37–1.55)	0%	>0.05
≥3 years	4	1.49(1.12–1.98)	91%	>0.05
Exposure frequency				
1 dispensation	5	0.96(0.72–1.26)	98%	>0.05
2 dispensations	5	1.36(1.03–1.78)	95%	>0.05
≥3 dispensations	10	2.02(1.49–2.74)	97%	>0.05
Types of antibiotics				
Cephalosporins	4	1.62(1.26–2.08)	83%	>0.05
Penicillin	4	1.62(1.26–2.08)	83%	>0.05
Macrolides	4	1.50(1.14–1.98)	87%	>0.05
Quinolones	5	1.49(1.45–1.54)	32%	>0.05
Tetracyclines	5	1.49(1.44 1.54)	47%	>0.05
Metronidazole and tinidazole	5	1.34(1.30–1.38)	46%	>0.05
Regional differences				
Developing countries	5	1.92(0.71–5.19)	98%	>0.05
Developed countries	19	1.58(1.27–1.96)	96%	>0.05

Quinolones, tetracyclines, and metronidazole and tinidazole were analyzed using a fixed effects model to calculate the pooled odds ratio (OR) values.

(Fig. S5). However, according to subgroup analysis results stratified by age at onset, there appears to be no significant association between antibiotic exposure and the incidence risk of children CD (OR, 1.26; 95% CI, 0.57–2.76) (Fig. S6).

Exposure period. We further analyzed the relationship between antibiotic exposure at different periods and the overall risk of IBD. The results show a significant correlation between antibiotic exposure during adulthood and the incidence risk of IBD (OR, 1.61; 95% CI, 1.22–2.11) (Fig. S7). Similarly, there is also a significant association between antibiotic exposure during childhood and the incidence risk of IBD (OR, 1.50; 95% CI, 1.03–2.18) (Fig. S8).

Year before diagnosis. Our meta-analysis comprehensively examined year of antibiotic exposure before the diagnosis of IBD. Our results indicate a positive association between antibiotic exposure of ≥3 years prior to the diagnosis of IBD and the incidence of IBD (OR, 1.49; 95% CI, 1.12–1.98). Similarly, there is a significant positive association between antibiotic exposure and the risk of developing IBD for patients with antibiotic exposure 2 years prior to diagnosis (OR, 1.46; 95% CI, 1.37–1.55) and ≤1 year prior to diagnosis (OR, 1.55; 95% CI, 1.17–2.04). (Fig. 4).

Exposure frequency. We conducted a rigorous analysis to investigate the correlation between varying frequencies of antibiotic exposure and the incidence of IBD. Cumulative frequency of antibiotic exposure from birth to the diagnosis of IBD in the included literature was collected for analysis. The association between cumulative frequency of antibiotic exposure and the risk of developing IBD was analyzed. The results showed that the cumulative frequency of antibiotic exposure with ≥3 dispensations had the greatest impact on the risk of developing IBD (OR, 2.02; 95% CI, 1.49–2.74). The frequency of exposure to two dispensations was also significantly associated with the risk of IBD (OR, 1.36; 95% CI, 1.03–1.78). However, the frequency of antibiotic exposure to a single dispensation was not significantly associated with the risk of developing IBD (OR, 0.96; 95% CI, 0.72–0.26) (Fig. 5). In addition, the dose–response curve showed a positive correlation between cumulative antibiotic exposure frequency and IBD risk (Fig. 6).

Types of antibiotics. We performed an in-depth analysis of the effect of exposure to various classes of antibiotics on IBD incidence. Our meta-analysis demonstrated a significant association between the use of specific antibiotics and the risk of developing IBD. Notably, exposure to cephalosporins was strongly linked to a positive association with the risk of new-onset IBD (OR, 1.62; 95% CI, 1.26–2.08), as was exposure to penicillins (OR, 1.50; 95% CI, 1.14–1.98) and macrolides (OR, 1.50; 95% CI, 1.14–1.98). In addition, exposure to quinolones (OR, 1.49; 95% CI, 1.45–1.54), tetracyclines (OR, 1.49; 95% CI, 1.44–1.54), and metronidazole and tinidazole (OR, 1.34; 95% CI, 1.30–1.38) was significantly associated with a positive association with the risk of IBD. These findings are detailed in Figure S9.

Regional differences. Our results revealed a significant association between antibiotic exposure and the risk of developing IBD in developed countries (OR, 1.58; 95% CI, 1.27–1.96). However, no significant correlation was observed in developing countries and some regions in China (OR, 1.92; 95% CI, 0.71–5.19), specifically Malaysia, Indonesia, and Yunnan, Guangzhou. This difference may be attributed to the limited number of studies in developing countries (5) and the larger number in developed countries (19). This disparity may be attributed to the limited number of studies conducted in developing countries (5) compared to the larger number in developed countries (19). Furthermore, in comparison to developed countries, developing nations have a shorter history of antibiotic usage, and the impact of antibiotic exposure on IBD may not have fully manifested yet (Fig. 7).

Discussion

In our meta-analysis, a total of 99 104 patients with IBD and 2 273 336 controls were included. Subgroup analyses were conducted based on year of antibiotic exposure before diagnosis, cumulative frequency of antibiotic exposure, specific types of antibiotics, and regional differences. These factors have not been addressed in previous meta-analyses. Firstly, we systematically reviewed and performed a meta-analysis of the included case–control and cohort studies. The findings indicated that antibiotic exposure is significantly positively associated with the risk of developing IBD.

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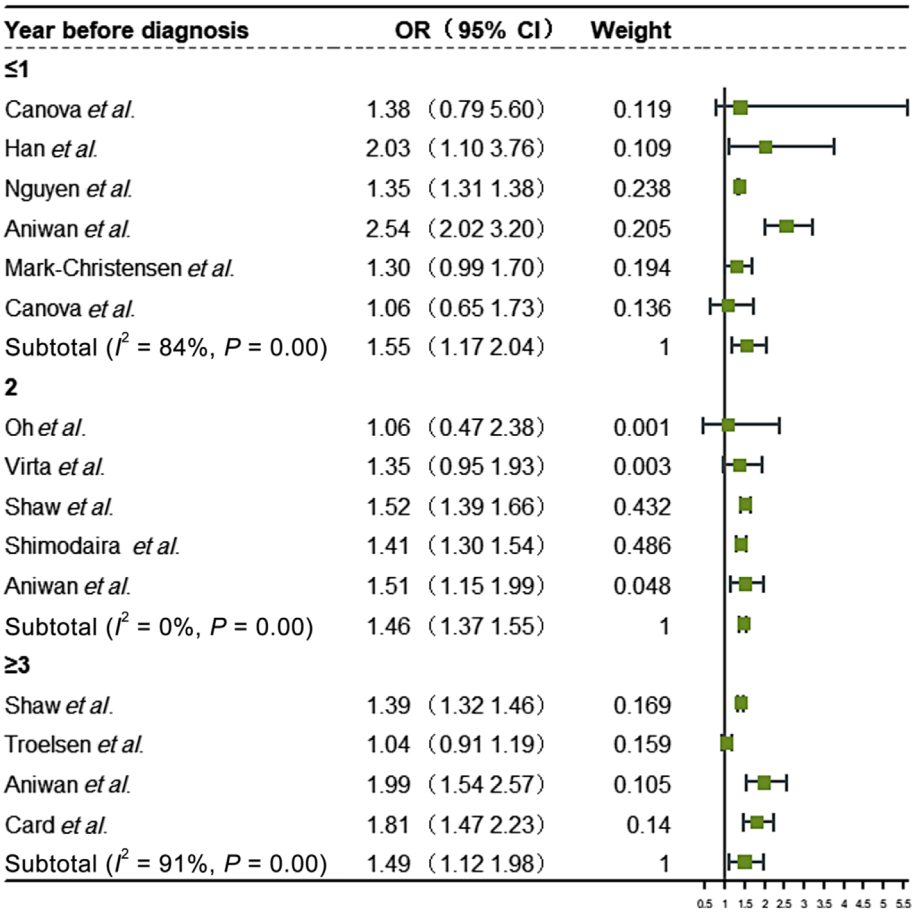


Figure 4 Forest plot of OR and 95% CI for the association between antibiotic exposure year before diagnosis and IBD incidence.

Furthermore, we analyzed the association between antibiotic exposure and the incidence of UC and CD in children and adults separately. The results revealed that antibiotic exposure significantly had a positive association with the risk of developing UC in adults; however, it had no significant effect on the incidence of UC or CD in children or CD in adults.

Analysis of the timing of antibiotic exposure and the risk of developing IBD showed that both childhood and adult antibiotic exposure is significantly positively associated with the risk of developing IBD, with no statistically significant difference between the two periods.

Subgroup analyses revealed a significant positive association between antibiotic exposure ≤1 year, 2 years, and ≥3 years prior to the IBD diagnosis and the risk of developing IBD. Furthermore, there was no observed dose–effect relationship between the duration of antibiotic exposure prior to diagnosis and the risk of developing IBD.

This meta-analysis also examined the association between cumulative frequency of antibiotic exposure from birth to diagnosis of IBD and the risk of developing IBD, showing that low-frequency antibiotic exposure (one dispensation) was not significantly associated with the risk of developing IBD. However, as the frequency of antibiotic exposure increased (≥2 dispensations), so did its effect on the risk of developing IBD.

The moderate use of antibiotics is essential in combating bacterial infections. However, research has indicated that the excessive use of antibiotics can lead to a reduction in the diversity of intestinal microbiota, a decrease in the secretion of antimicrobial peptides, and an increase in bacterial resistance. This can make the human body more susceptible to intestinal diseases.[33] These findings suggest a close correlation between gut microbiota and IBD risk loci, with dysbiosis of the gut microbiota potentially contributing to the development of IBD.[34]

Our meta-analysis offered valuable insights into the correlation between specific types of antibiotics and the risk of IBD. The study results revealed that most antibiotics are significantly positively associated with the likelihood of developing IBD, with penicillin and cephalosporins having the most significant impact on IBD risk. In contrast, metronidazole and tinidazole had relatively minor effects. This variance may be attributed to the historical usage of these antibiotics. Considering that penicillin and cephalosporins were introduced in 1928 and 1945, respectively, they have been in use for nearly 80–100 years. On the other hand, metronidazole and tinidazole have been utilized in gingivitis treatment

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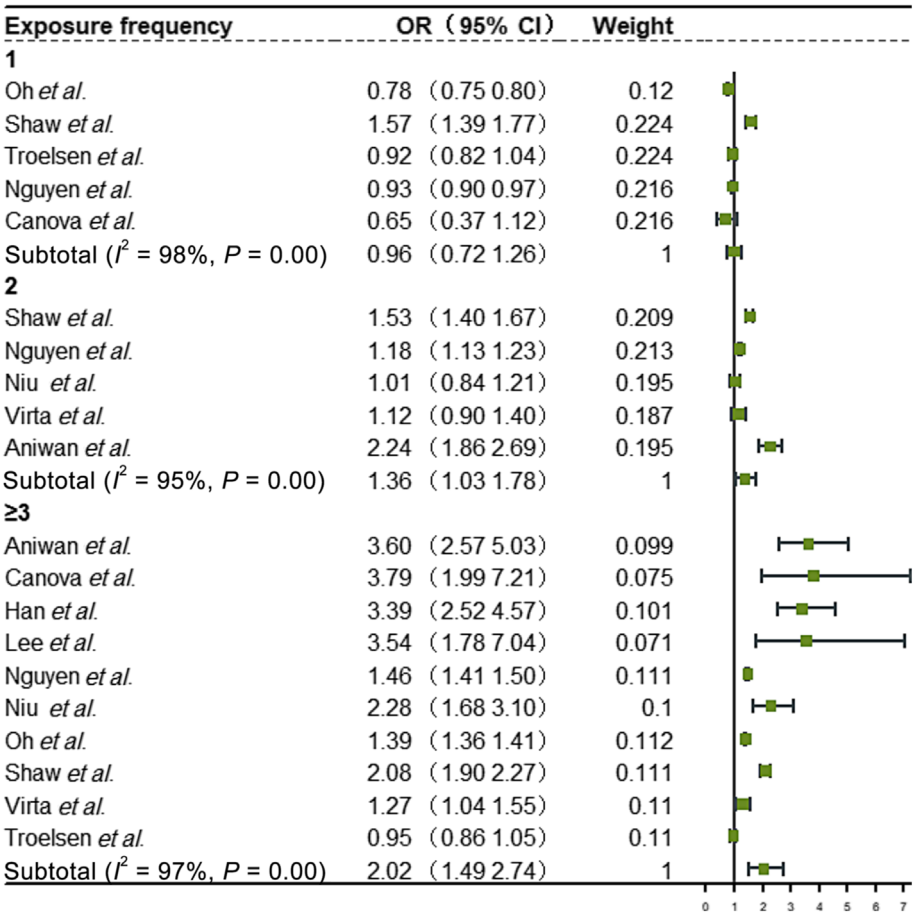


Figure 5 Forest plot of OR and 95% CI for association between antibiotics exposure frequency and IBD incidence.

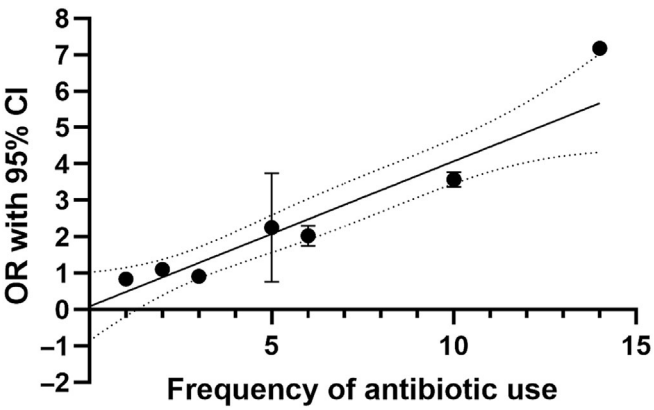


Figure 6 Dose–response relationship between cumulative antibiotic exposure frequency and the risk of new-onset IBD.

since 1962. However, it is concerning that exposure to these relatively new antibiotics can also potentially affect the incidence of IBD over time.

Another interesting finding of this study was that in developed countries, antibiotic exposure is significantly positively associated with the risk of developing IBD. However, there was no significant relationship between antibiotic exposure and IBD incidence in developing countries. This is similar to the characteristics of IBD that have long been prevalent mainly in developed countries. We speculate that this difference may be due to a longer history of antibiotic use in developed countries than in developing countries, where the effects of antibiotic use on IBD are not yet fully apparent. In addition, the limited sample size of research literature in developing countries may also be a contributing factor.

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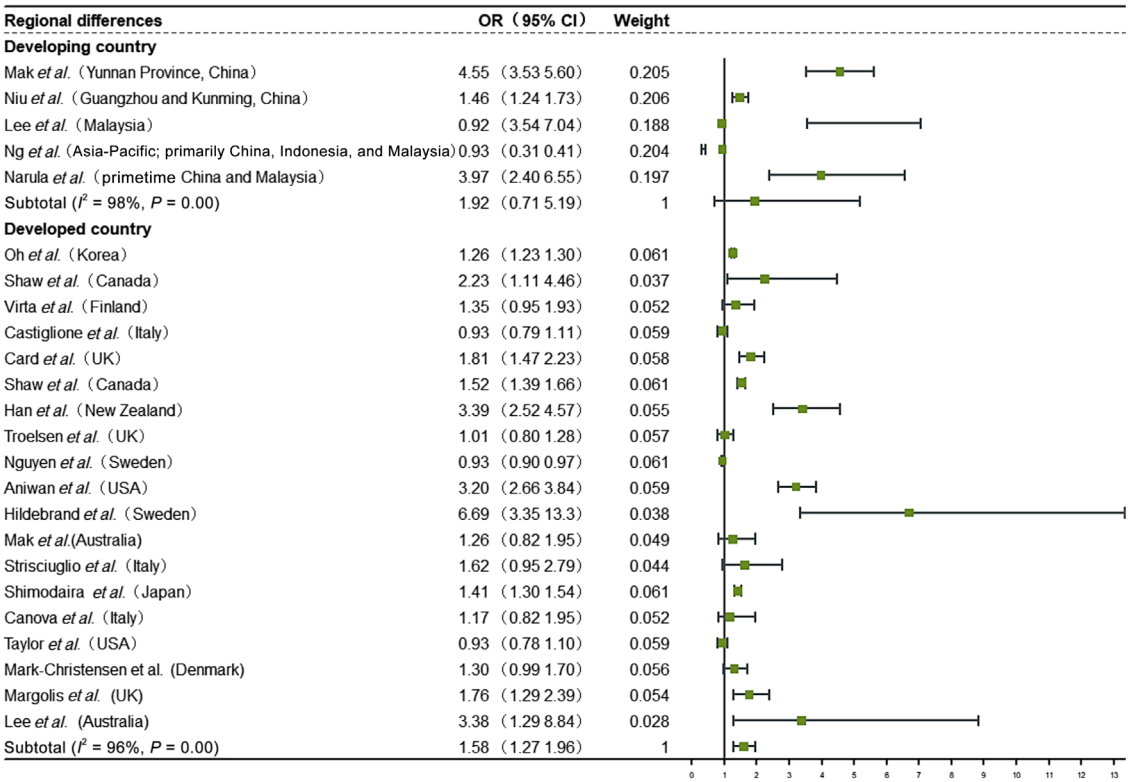


Figure 7 Forest plot of OR and 95% CI for association between the location of patients and the incidence of IBD.

A study by Dar *et al*.[35] revealed a significant correlation between antibiotic exposure and the risk of CD in children. They included 29 880 IBD patients and 715 548 controls. In contrast, our meta-analysis did not observe a significant positive association between antibiotic exposure and the risk of CD incidence in children. This difference can be attributed to the broader and more comprehensive sample size included in our analysis, as well as differences in analysis methods. The study of Ruqiao *et al*.[36] showed that there was an association between exposure to metronidazole drugs and a higher risk of IBD. However, our findings suggest that metronidazole antibiotics have a modest effect on IBD incidence. Further research is needed on the association of different antibiotics with the risk of developing IBD.

Conclusions

This meta-analysis revealed a significant association between antibiotic exposure and a positive association with the risk of IBD. The study found that the more frequently antibiotics were used from birth to the diagnosis of IBD, the higher the risk of developing the condition. Furthermore, it was observed that most types of antibiotics, particularly penicillin and cephalosporins, are linked to an elevated risk of IBD. Interestingly, this association between antibiotic exposure and IBD risk was only evident in developed countries and not in developing countries. Therefore, it is recommended that unnecessary antibiotic use be reduced and measures to strengthen antibiotic management be implemented as effective strategies for reducing the prevalence of IBD.

References

1 Le Berre C, Honap S, Peyrin-Biroulet L. Ulcerative colitis. *Lancet* 2023; **402**: 571–84. https://doi.org/10.1016/s0140-6736(23)00966-2 PubMed PMID: 37573077.
2 Li J, Butcher J, Mack D, Stintzi A. Functional impacts of the intestinal microbiome in the pathogenesis of inflammatory bowel disease. *Inflamm. Bowel Dis.* 2015; **21**: 139–53. https://doi.org/10.1097/mib.0000000000000215 PubMed PMID: 25248007.
3 Ianiro G, Tilg H, Gasbarrini A. Antibiotics as deep modulators of gut microbiota: between good and evil. *Gut* 2016; **65**: 1906–15. Epub 20160816. https://doi.org/10.1136/gutjnl-2016-312297 PubMed PMID: 27531828.
4 Pérez-Cobas AE, Gosalbes MJ, Friedrichs A *et al*. Gut microbiota disturbance during antibiotic therapy: a multi-omic approach. *Gut* 2013; **62**: 1591–601. Epub 20121212. https://doi.org/10.1136/gutjnl-2012-303184 PubMed PMID: 23236009; PubMed Central PMCID: PMC3812899.
5 Ungaro R, Bernstein CN, Gearry R *et al*. Antibiotics associated with increased risk of new-onset Crohn's disease but not ulcerative colitis: a meta-analysis. *Am. J. Gastroenterol.* 2014; **109**: 1728–38. Epub 20140916. https://doi.org/10.1038/ajg.2014.246 PubMed PMID: 25223575.
6 Yu-Chen H, Hsin N, Jessie H *et al*. Research trends in technology-based learning from 2000 to 2009: a content analysis of publications in selected journals. *J. Educ. Technol. Soc.* 2012; **15**: 354–70.
7 Hwang GJ, Tsai CC. Research trends in mobile and ubiquitous learning: a review of publications in selected journals from 2001 to 2010. *J. Educ. Technol.* 2011; **42**: E65–70. https://doi.org/10.1111/j.1467-8535.2011.01183.x.

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8 Hutton B, Salanti G, Caldwell DM *et al*. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. *Ann. Intern. Med.* 2015; **162**: 777–84. https://doi.org/10.7326/m14-2385 PubMed PMID: 26030634.

9 Wells GA, Shea BJ, O'Connell D, Peterson J, Tugwell P. The Newcastle–Ottawa scale (NOS) for assessing the quality of non-randomized studies in meta-analysis. 2023.

10 Oh SJ, Kim HJ, Lee CK. A dose-dependent increase in the risk of inflammatory bowel disease after exposure to broad-spectrum antibiotics: a national population study in Korea. *Aliment. Pharmacol. Ther.* 2023; **58**: 191–206. Epub 20230508. https://doi.org/10.1111/apt.17542 PubMed PMID: 37154240.

11 Shimodaira Y, Watanabe K, Iijima K. The risk of antibiotics and enterocolitis for the development of inflammatory bowel disease: a Japanese administrative database analysis. *Sci. Rep.* 2022; **12**: 7604. Epub 20220509. https://doi.org/10.1038/s41598-022-11646-2 PubMed PMID: 35534662; PubMed Central PMCID: PMC9085770.

12 Mak JWY, Yang S, Stanley A *et al*. Childhood antibiotics as a risk factor for Crohn's disease: the ENIGMA international cohort study. *J. G. H. Open* 2022; **6**: 369–77. Epub 20220601. https://doi.org/10.1002/jgh3.12755 PubMed PMID: 35774350; PubMed Central PMCID: PMC9218523.

13 Lee WS, Song ZL, Wong SY *et al*. Environmental risk factors for inflammatory bowel disease: a case control study in Southeast Asian children. *J. Paediatr. Child Health* 2022; **58**: 782–90. Epub 20211111. https://doi.org/10.1111/jpc.15830 PubMed PMID: 34761458.

14 Nguyen LH, Örtqvist AK, Cao Y *et al*. Antibiotic use and the development of inflammatory bowel disease: a national case-control study in Sweden. *Lancet Gastroenterol. Hepatol.* 2020; **5**: 986–95. Epub 20200817. https://doi.org/10.1016/s2468-1253(20)30267-3 PubMed PMID: 32818437; PubMed Central PMCID: PMC8034612.

15 Canova C, Ludvigsson JF, Di Domenicantonio R, Zanier L, Barbiellini Amidei C, Zingone F. Perinatal and antibiotic exposures and the risk of developing childhood-onset inflammatory bowel disease: a nested case-control study based on a population-based birth cohort. *Int. J. Environ. Res. Public Health* 2020; **17**(). Epub 20200402. https://doi.org/10.3390/ijerph17072409 PubMed PMID: 32252276; PubMed Central PMCID: PMC7177699.

16 Troelsen FS, Jick S. Antibiotic use in childhood and adolescence and risk of inflammatory bowel disease: a case-control study in the UK clinical practice research datalink. *Inflamm. Bowel Dis.* 2020; **26**: 440–7. https://doi.org/10.1093/ibd/izz137 PubMed PMID: 31265060.

17 Aniwan S, Tremaine WJ, Raffals LE, Kane SV, Loftus EV Jr. Antibiotic use and new-onset inflammatory bowel disease in Olmsted County, Minnesota: a population-based case-control study. *J. Crohns Colitis* 2018; **12**: 137–44. https://doi.org/10.1093/ecco-jcc/jjx135 PubMed PMID: 29029150; PubMed Central PMCID: PMC5881749.

18 Strisciuglio C, Giugliano F, Martinelli M, Cenni S, Greco L, Staiano A, Miele E. Impact of environmental and familial factors in a cohort of pediatric patients with inflammatory bowel disease. *J. Pediatr. Gastroenterol. Nutr.* 2017; **64**: 569–74. https://doi.org/10.1097/mpg.0000000000001297 PubMed PMID: 27306105.

19 Niu J, Miao J, Tang Y *et al*. Identification of environmental factors associated with inflammatory bowel disease in a southwestern highland region of China: a nested case-control study. *PLoS ONE* 2016; **11**: e0153524. Epub 20160412. https://doi.org/10.1371/journal.pone.0153524 PubMed PMID: 27070313; PubMed Central PMCID: PMC4829194.

20 Ng SC, Tang W, Leong RW *et al*. Environmental risk factors in inflammatory bowel disease: a population-based case-control study in Asia-Pacific. *Gut* 2015; **64**: 1063–71. Epub 20140912. https://doi.org/10.1136/gutjnl-2014-307410 PubMed PMID: 25217388.

21 Virta L, Auvinen A, Helenius H, Huovinen P, Kolho KL. Association of repeated exposure to antibiotics with the development of pediatric Crohn's disease--a nationwide, register-based Finnish case-control study. *Am. J. Epidemiol.* 2012; **175**: 775–84. Epub 20120224. https://doi.org/10.1093/aje/kwr400 PubMed PMID: 22366379.

22 Castiglione F, Diaferia M, Morace F *et al*. Risk factors for inflammatory bowel diseases according to the "hygiene hypothesis": a case-control, multi-centre, prospective study in southern Italy. *J. Crohns Colitis* 2012; **6**: 324–9. Epub 20111007. https://doi.org/10.1016/j.crohns.2011.09.003 PubMed PMID: 22405169.

23 Shaw SY, Blanchard JF, Bernstein CN. Association between the use of antibiotics and new diagnoses of Crohn's disease and ulcerative colitis. *Am. J. Gastroenterol.* 2011; **106**: 2133–42. Epub 20110913. https://doi.org/10.1038/ajg.2011.304 PubMed PMID: 21912437.

24 Shaw SY, Blanchard JF, Bernstein CN. Association between the use of antibiotics in the first year of life and pediatric inflammatory bowel disease. *Am. J. Gastroenterol.* 2010; **105**: 2687–92. Epub 20101012. https://doi.org/10.1038/ajg.2010.398 PubMed PMID: 20940708.

25 Han DY, Fraser AG, Dryland P, Ferguson LR. Environmental factors in the development of chronic inflammation: a case-control study on risk factors for Crohn's disease within New Zealand. *Mutat. Res.* 2010; **690**: 116–22. Epub 20090912. https://doi.org/10.1016/j.mrfmmm.2009.09.002 PubMed PMID: 19751746.

26 Hildebrand H, Malmborg P, Askling J, Ekbom A, Montgomery SM. Early-life exposures associated with antibiotic use and risk of subsequent Crohn's disease. *Scand. J. Gastroenterol.* 2008; **43**: 961–6. https://doi.org/10.1080/00365520801971736 PubMed PMID: 19086166.

27 Card T, Logan RF, Rodrigues LC, Wheeler JG. Antibiotic use and the development of Crohn's disease. *Gut* 2004; **53**: 246–50. https://doi.org/10.1136/gut.2003.025239 PubMed PMID: 14724158; PubMed Central PMCID: PMC1774910.

28 Taylor MT, Margolis DJ, Kwatra SG, Barbieri JS. A propensity score matched cohort study identifying an association of acne, but not oral antibiotic or isotretinoin use, with risk of incident inflammatory bowel disease. *J. Am. Acad. Dermatol.* 2023; **88**: 841–7. Epub 20230120. https://doi.org/10.1016/j.jaad.2023.01.014 PubMed PMID: 36682724; PubMed Central PMCID: PMC10033360.

29 Narula N, Wong ECL, Pray C *et al*. Associations of antibiotics, hormonal therapies, oral contraceptives, and long-term NSAIDS with inflammatory bowel disease: results from the prospective urban rural epidemiology (PURE) study. *Clin. Gastroenterol. Hepatol.* 2023; **21**: 2649–59. Epub 20221215. https://doi.org/10.1016/j.cgh.2022.11.037 PubMed PMID: 36528284.

30 Mark-Christensen A, Lange A, Erichsen R, Frøslev T, Esen B, Sørensen HT, Kappelman MD. Early-life exposure to antibiotics and risk for Crohn's disease: a nationwide Danish birth cohort study. *Inflamm. Bowel Dis.* 2022; **28**: 415–22. https://doi.org/10.1093/ibd/izab085 PubMed PMID: 34000050; PubMed Central PMCID: PMC8889299.

31 Lee TW, Russell L, Deng M, Gibson PR. Association of doxycycline use with the development of gastroenteritis, irritable bowel syndrome and inflammatory bowel disease in Australians deployed abroad. *Intern. Med. J.* 2013; **43**: 919–26. https://doi.org/10.1111/imj.12179 PubMed PMID: 23656210.

32 Margolis DJ, Fanelli M, Hoffstad O, Lewis JD. Potential association between the oral tetracycline class of antimicrobials used to treat acne and inflammatory bowel disease. *Am. J. Gastroenterol.* 2010; **105**: 2610–6. Epub 20100810. https://doi.org/10.1038/ajg.2010.303 PubMed PMID: 20700115.

33 Santacroce L, Di Domenico M, Montagnani M, Jirillo E. Antibiotic resistance and microbiota response. *Curr. Pharm. Des.* 2023; **29**: 356–64. https://doi.org/10.2174/1381612829666221219093450 PubMed PMID: 36537602.

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34 Eom T, Kim YS, Choi CH, Sadowsky MJ, Unno T. Current understanding of microbiota- and dietary-therapies for treating inflammatory bowel disease. *J. Microbiol.* 2018; **56**: 189–98. Epub 20180228. https://doi.org/10.1007/s12275-018-8049-8 PubMed PMID: 29492876.
35 Dar SH, Maniya MT, Merza N *et al*. The association of antibiotic exposure with new-onset inflammatory bowel disease: a systematic review and meta-analysis. *Clin. Res. Hepatol. Gastroenterol.* 2023; **47**: 102129. Epub 20230426. https://doi.org/10.1016/j.clinre.2023.102129 PubMed PMID: 37116651.
36 Duan R, Zhang C, Li G, Li J, Duan L. Antibiotic exposure and risk of new-onset inflammatory bowel disease: a systematic review and dose-response meta-analysis. *Clin. Gastroenterol. Hepatol.* 2024. Epub 20240228. https://doi.org/10.1016/j.cgh.2024.02.010 PubMed PMID: 38423349.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1. Literature selection process.
Figure S2. Forest plot of ORs and 95% CIs for association between antibiotic exposure and risk of IBD when excluding each study individually for prescription-based studies.
Figure S3. The forest plot summarizes the results of the meta-analysis on adult ulcerative colitis (UC).
Figure S4. The forest plot summarizes the results of the meta-analysis on children ulcerative colitis (UC).
Figure S5. The forest plot summarizes the results of the meta-analysis on adult Crohn's disease (CD).
Figure S6. The forest plot summarizes the results of the meta-analysis on children Crohn's disease (CD).
Figure S7. The forest plot summarizes the results of the meta-analysis on adult Inflammatory bowel disease (IBD).
Figure S8. The forest plot summarizes the results of the meta-analysis on children Inflammatory bowel disease (IBD).
Figure S9. Forest plot of OR and 95% CI for association between specific types of antibiotics and IBD incidence.
Table S1. Details of the reviewed studies.

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